

Abstract

Information spread refers to the way in which information moves throughout a population. When looking at information spread on online networks specifically, this happens very quickly and can be influenced by a number of factors. The goal of modeling is to incorporate these factors so that we can create a model that gives an accurate picture of the way that information spreads online. There are multiple types of mathematical models that have been used to show this phenomenon. One of the most common is the basic SIR model, where information is analogous to an infectious disease. In order to better define the classes in terms of information spread, the classes will be changed to Unaware, Informed, and Disinterested. This model will serve as a comparison for more complex models. The UID model is expanded to include a reinfection rate, which allows users to move back and forth between the Informed and Disinterested classes. We perform equilibria analysis and run simulations to determine the long-term behavior of the system. Another model that is considered is an absorbing Markov chain, which is based off of the modified forest-fire model. The behavior of this matrix is analyzed through simulations and the steady-state and fundamental matrices. The results of these two models are compared, along with those from the basic model, by looking at the predicted population proportions in each class after a 24 hour time period in order to determine which factors affect the way in which information spreads online.

Introduction

The way that information spreads on social networks is a topic that is growing in popularity today. People are very highly interconnected online and any piece of information has the ability to spread through the network at a high speed. Since there are hundreds of millions of

users on any given social network, the information can easily reach a large portion of the population, even if it begins with only a small number. There are many different factors that affect how information spreads and therefore how many users receive it. For example, the number of people interested in a piece of information changes depending on the topic, so some things may spread much faster or become more widespread than others. Also, recent information is generally more interesting and engaging for the population, so that type of information spreads faster as well. By modeling how this information spread happens, we have the ability to better understand the network's users and their behavior.

One of the most common ways that information spread is modeled is using the traditional SIR (Susceptible-Infected-Recovered) model for infectious disease spread. In this sense, the information is thought of as a “disease” that is spreading through the population. Then, the susceptible class represents a person that doesn't know the information, the infected class represents a person that knows about the information and is spreading it to others, and the recovered class represents a person that is no longer interested in the information. This model uses differential equations to show how the proportion of the population in each class changes over time. While this model is useful as a starting point, it can also be expanded in order to more accurately reflect the reality of the way information spreads online.

In a similar model that was created to look at the spread of memes online, researchers included a reinfection rate to demonstrate how users that “recovered” could become re-interested in the information later (Wang). This modification allows for more complexity within the model. In real-world scenarios, it is often the case that there are multiple spikes in interest in a topic over a period of time, so it makes sense that the model would include becoming re-interested as an option. This model can be compared to the basic UID model and therefore used to answer

questions about the factors that affect information spread. For example, how many users know about the information after a 24 hour time period? In the long term, what proportion of the population is informed? Does the inclusion of a reinfection rate significantly affect either of these values?

A third type of model that has been created to explore information spread is known as the modified forest-fire model. This model distinguishes between spreaders and non-spreaders on the network (Kumar). There are a large number of people that see information online and choose not to share it further, for whatever reason, so the number of users that spread the information may not actually represent the proportion of the population that is informed. The model includes four classes, where the “Fire” state represents a spreader of information and the “Burnt” state represents a non-spreader of information (Kumar). The combination of these two proportions may give a better estimate of the true proportion of the population that is informed. The results from this model can again be compared with the models above to determine how the idea of a “Burnt” state changes the predicted spread of information.

Model Development and Assumptions

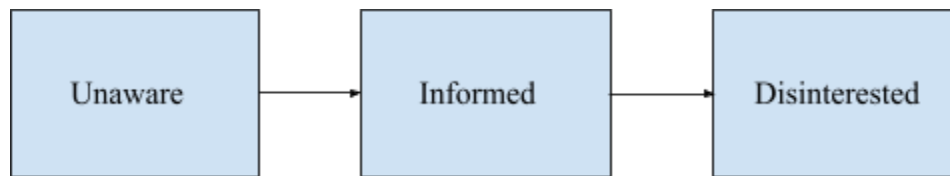
The classic Susceptible-Infected-Recovered (SIR) model divides the population into three groups. While it is usually used in terms of infectious disease spread, these categories can also be applied to information spread online, so the titles of the classes will become Unaware, Informed, and Disinterested. The unaware class refers to individuals that do not know the information, the informed class refers to individuals that know the information and are spreading it to others, and the disinterested class refers to individuals that are no longer spreading the information. The equations for this model are as follows:

$$\frac{dU}{dt} = -\alpha UI$$

$$\frac{dI}{dt} = \alpha UI - \beta I$$

$$\frac{dD}{dt} = \beta I$$

The parameter α represents the transmission coefficient, meaning the rate at which people who do not know the information learn about it. The parameter β represents the recovery rate, meaning the rate at which people who have the information stop spreading it (Smith). This model assumes that the parameters α and β do not change, so the proportion of the population that becomes informed or disinterested stays the same each day (Smith). Also, we assume that once a person stops spreading the information, or becomes “disinterested”, they are no longer capable of moving back to the informed state. These dynamics are shown in the compartmental diagram below.



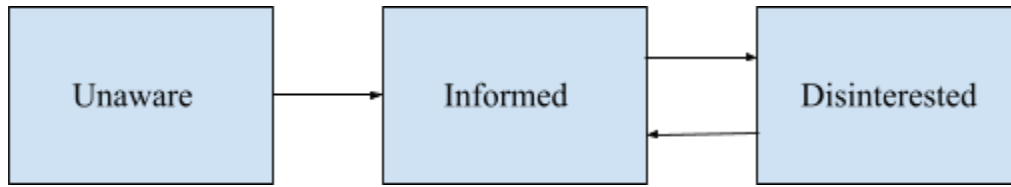
The UID model above will be compared with another that includes the possibility of becoming re-interested in the information. This will reflect the complexity of information spread that is seen online. For example, users may lose and regain interest in specific topics over a period of time, like during a sports season or throughout an election cycle. The population is still divided into three classes (in this case: unaware, informed, and disinterested), but because there is a reinfection rate, the disinterested individuals are able to be informed again later (Wang). The equations for this model are as follows:

$$\frac{dU}{dt} = -\alpha UI$$

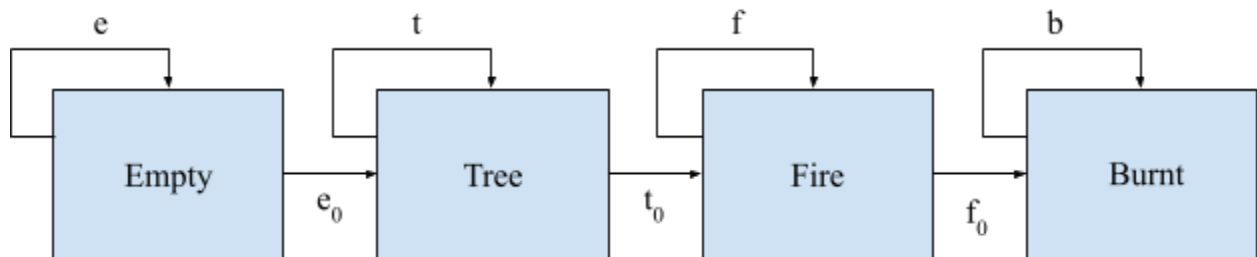
$$\frac{dI}{dt} = \alpha UI + \gamma ID - \beta I$$

$$\frac{dD}{dt} = -\gamma ID + \beta I$$

In this case, the parameter α still represents the transmission rate and β represents the recovery rate, but γ represents the reinfection rate. The transitions between classes are shown below.



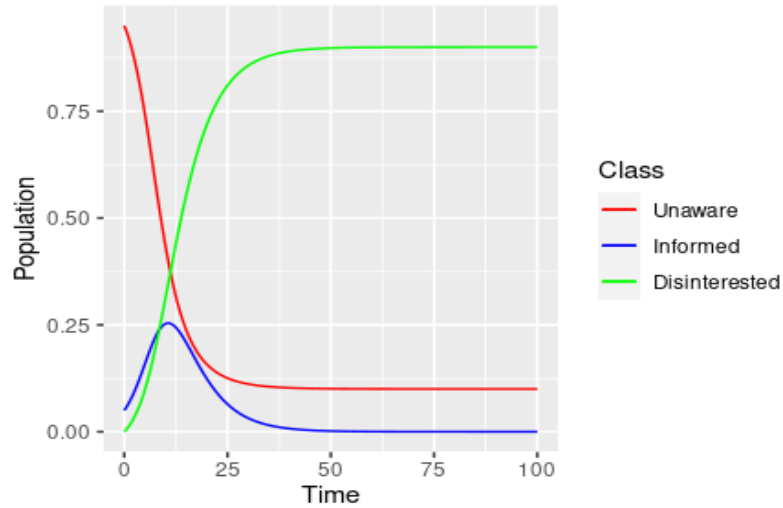
Finally, these two models will be compared to the modified forest-fire model. This type of model is unique because it distinguishes between individuals who spread the information and those who do not (Kumar). The population is divided into four classes: Empty, Tree, Fire, and Burnt. Empty represents a user that isn't on the network at all, Tree represents a user that is online, but doesn't know about the information, Fire represents a user that has the information and spreads it, and Burnt represents a user that has the information, but is no longer spreading it (Kumar). The original model uses an algorithm and calculates probabilities for each individual user, but we adapt these dynamics into a Markov Chain matrix model. The possible transitions between states are shown below:



Since this is a matrix model, the parameter values all represent the probabilities of moving between states. The parameter e represents the probability of a user that is in Empty staying there, meaning that they remain not on the network. The parameter e_0 represents the probability of a user moving from Empty to Tree, meaning that they join the network. The parameter t represents the probability of a user staying in Tree, meaning that they remain uninformed, while t_0 represents a move from Tree to Fire, meaning that they learn about the information. The parameter f represents a user that remains in Fire, so they are still talking about the information and spreading it to others. The parameter f_0 represents the probability that a user moves from Fire to Burnt, meaning that they stop spreading the information. In this model, the Burnt state is an absorption state, so once a user enters that state, they stay there. Therefore, the parameter b is equal to 1.

Analysis and Results

The basic version of the UID model will be used as a comparison for the other, more complex models. Since this model only allows for users to move from the Unaware state, to the Informed state, and then to Disinterested state, there will be less focus on equilibria analysis, and instead we will move into running simulations. The parameters chosen for this model are $\alpha = 0.5$ and $\beta = 0.2$ and the initial conditions are $U_0 = 0.95$, $I_0 = 0.05$, and $D_0 = 0$. This means that we will be looking at population proportions rather than actual population values. The transmission and recovery rates, α and β , may seem high, but these values were chosen in order to be more similar to the higher probabilities of moving between states that are seen in the matrix model, which will be discussed in more detail later.



The graph above does stabilize with specific proportions of the population in each class in the long term. There is 10% of the population in the Unaware class, 0% of the population in the Informed class, and 90% of the population in the Disinterested class. In order to compare with the matrix model later, we will also consider the population proportions after 24 hours. In this case, there is 13% of the population in the Unaware class, 7.4% in the Informed class, and 79.6% in the Disinterested class.

Next, the UID model with a reinfection rate was analyzed to determine long-term behavior. In order to simplify the analysis, the population size, N , is assumed to be a constant, meaning that the third category of the model can be calculated by $D = N - U - I$. This allows the third equation to be removed from the system, so the model becomes:

$$\frac{dU}{dt} = -\alpha UI$$

$$\frac{dI}{dt} = (\alpha - \gamma)UI + I(\gamma N - \gamma I - \beta)$$

The equilibria found for this model are $(0,0)$, $(0, N - \frac{\beta}{\gamma})$, and $(\frac{\gamma N - \beta}{\gamma - \alpha}, 0)$. The second equilibrium, $(0, N - \frac{\beta}{\gamma})$, deals with the case where the information spread persists, which is the most interesting case, so this is where we will focus our analysis. Since the model deals with

population values, they cannot be negative, which creates some restrictions on the parameters.

First, $N - \frac{\beta}{\gamma} > 0$ must be true for the second equilibria, so $\frac{N\gamma}{\beta} > 1$. This means that $\frac{N\gamma}{\beta}$ is the basic reproduction number of the system, known as R_0 .

The R_0 value represents the transmissibility of contagiousness or a disease, or in this case, information on a social network (Delamater). In order for the disease to continue spreading, R_0 must be greater than 1. Since this equilibrium predicts continued information spread, it makes sense that R_0 will be some value greater than 1, but it can actually be calculated using parameter values. These will be discussed more later, but assuming that $\beta = 0.05$, $\gamma = 0.1$, and $N=1$, we find R_0 to be 2. In the context of infectious diseases, this means that there is expected to be 2 secondary cases from a single infection (Delamater). However, for the purposes of information spread, this means that for each person that knows the information, they are expected to spread it to two others. Initially, this doesn't seem like a very high number, but once two other users have the information, they will in turn spread it to four people, then eight more, and so on. This allows the information to actually spread through the population quite quickly.

The stability of the equilibria can be determined using the Jacobian matrix. The Jacobian for this system is:

$$J = \begin{bmatrix} -\alpha I & -\alpha U \\ \alpha I - \gamma I & \alpha U + \gamma(N - U - I) - 2\gamma I - \beta \end{bmatrix}$$

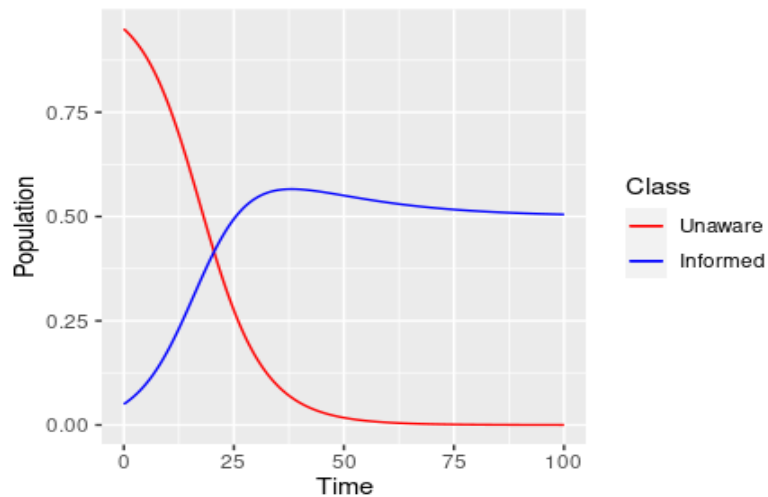
This matrix can be evaluated for the equilibria and the eigenvalues can be used in order to determine the stability. The Jacobian for the equilibrium $(0, N - \frac{\beta}{\gamma})$ is:

$$J(0, N - \beta\gamma) = \begin{bmatrix} \frac{-\alpha}{\gamma}(\gamma N - \beta) & 0 \\ \frac{(\gamma N - \beta)(\alpha - \gamma)}{\gamma} & -2(\gamma N - \beta) \end{bmatrix}$$

Since the matrix is triangular, the eigenvalues are easily found to be $\frac{-\alpha}{\gamma}(\gamma N - \beta)$ and $-2(\gamma N - \beta)$.

Based on the conditions for the equilibrium, we know that $\gamma N - \beta > 0$, so both of the eigenvalues will be negative. This means that the equilibrium is a sink, which is stable.

Another way to analyze this model's behavior is to run simulations. We assume parameter values of $\alpha = 0.2$, $\beta = 0.05$, and $\gamma = 0.1$. The population size remains constant, so $N = 1$, and we will assume that 95% of the population starts out being unaware of the information, so $U_0 = 0.95$ and $I_0 = 0.05$. After running the simulation, the population proportions stabilize at $U = 0$ and $I = 0.5$, which is equivalent to the equilibrium $(0, N - \frac{\beta}{\gamma})$, once parameter values are substituted.



Since we made the substitution of $D = N - U - I$, we can also determine that the long-term proportion of the population in the disinterested class is $D = 0.5$. The addition of the reinfection rate changes the long-term population proportions of the model. In this model, a larger proportion of the population remains in the informed state because they are able to become re-interested, which isn't a possibility in the basic UID model. In order to better compare the results of the modified UID model with the following matrix model, we will look at the predicted

proportions after 24 hours. This model predicts that 30.3% of the population will be unaware, 47.9% of the population will be informed, and 21.8% of the population will be disinterested in the information.

The matrix model can also be analyzed in order to determine the long-term population proportions. These results will be compared to those of the two models above. The transition matrix can be formulated based off of the compartmental diagram and the result is shown below.

$$T = \begin{bmatrix} e & 0 & 0 & 0 \\ e_0 & t & 0 & 0 \\ 0 & t_0 & f & 0 \\ 0 & 0 & f_0 & 1 \end{bmatrix}$$

The first column represents the Empty state, the second column represents the Tree state, the third column represents the Fire state, and the fourth column represents the Burnt state. This matrix can be decomposed into the following pieces:

$$A = \begin{bmatrix} e & 0 & 0 \\ e_0 & t & 0 \\ 0 & t_0 & f \end{bmatrix}, B = [0 \quad 0 \quad f_0], O = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, I = [1]$$

Each of these matrices is important in determining the steady-state matrix, L, for the model and the fundamental matrix, F. The L matrix represents the proportion of the population in each state in the long term. The lower left corner of the matrix contains the absorption probabilities, meaning the chance that the system will be absorbed given that it started in a specific state. Since there is only one absorption state in this model, each probability in this section will end up being equal to 1. The general form of the steady-state matrix is:

$$L = \begin{bmatrix} O_{3 \times 3} & O_{3 \times 1} \\ B(I - A)^{-1} & I_{1 \times 1} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \frac{-e_0 f_0 t_0}{(e-1)(f-1)(t-1)} & \frac{f_0 t_0}{(f-1)(t-1)} & \frac{f_0}{1-f} & 1 \end{bmatrix}$$

Another matrix that we can analyze is the fundamental matrix. This gives information about the number of time steps that the system is in a specific state and also the average number of time steps until the system is absorbed. The general form of this matrix is:

$$F = (I - A)^{-1} = \begin{bmatrix} \frac{1}{1-e} & 0 & 0 \\ \frac{e_0}{(e-1)(t-1)} & \frac{1}{1-t} & 0 \\ \frac{-e_0 t_0}{(e-1)(f-1)(t-1)} & \frac{t_0}{(f-1)(t-1)} & \frac{1}{1-f} \end{bmatrix}$$

Each individual entry f_{ij} represents the number of time steps that the system is in state j given that it began in state i . For example, the value $\frac{1}{1-t}$ means that the system will be in the Tree state for $\frac{1}{1-t}$ time steps, given that it also started in the Tree state. Also, the sum of the entries in each column represents the average number of time steps for the system to reach the absorption state from its initial state. In this case, the sum of the first column would be the average number of time steps for the system to reach the Burnt state if it initially started in the Empty state.

To make these interpretations more concrete and run simulations, we can substitute parameter values into the generalized matrices and evaluate them. The parameter values are based off of those in the modified forest-fire algorithm (Kumar). We will assume that the probabilities and initial conditions for the model are as follows:

Parameter/Initial Condition	Value	Description
e	0.8	Probability of staying in Empty state
e_0	0.2	Probability of moving from Empty to Tree
t	0.5	Probability of staying in Tree state
t_0	0.5	Probability of moving from Tree to Fire
f	0.5	Probability of staying in Fire state
f_0	0.5	Probability of moving from Fire to Burnt

E_0	70	Initial number of users in Empty
T_0	20	Initial number of users in Tree
F_0	10	Initial number of users in Fire
B_0	0	Initial number of users in Burnt

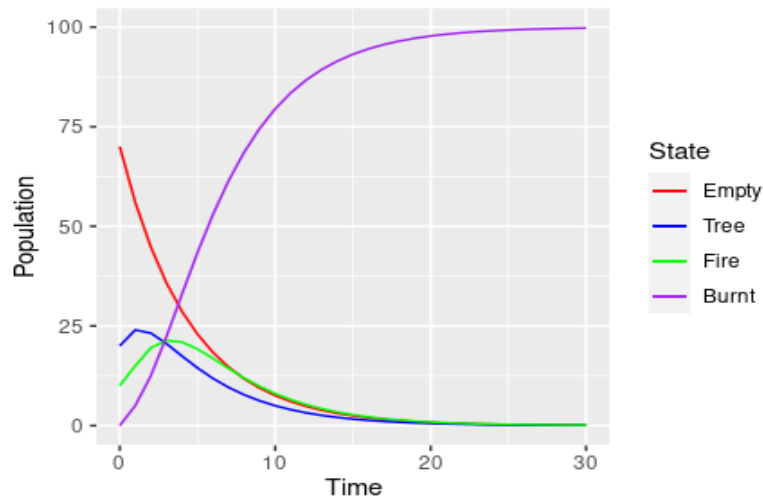
Now, with these values, the L matrix and the F matrix become:

$$L = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

$$F = \begin{bmatrix} 5 & 0 & 0 \\ 2 & 2 & 0 \\ 2 & 2 & 2 \end{bmatrix}$$

The absorption probabilities in the L matrix mean that the Empty, Tree, and Burnt states have a 100% chance of reaching the Burnt state. Since that is the only absorption state, this makes sense in the context of the model. The individual values in the F matrix represent the average number of time steps that the model is in that state, given that it began there. For example, the value 5 means that a user will be in the Empty state for about 5 hours, given that they started in Empty. The sums of each of the columns in F are also important. They represent the number of time steps that it takes for the system to reach the absorption state given that it started in a certain state. For example, the sum of the first column is 9, which means that it will take about 9 hours for a user initially in the Empty state to reach the Burnt state. The sum of the second column is 4, which means that it will take about 4 hours for a user initially in the Tree state to reach the Burnt state. Finally, the sum of the third column is 2, which means that it will take about 2 hours for a user initially in the Fire state to reach the Burnt state.

Similarly to the other two models, we can run simulations to see both the long term behavior of the system and the results after a 24 hour time period. Since this model includes the Burnt absorption state, in the long term, every user will end up there, but it is still interesting to look at the dynamics.



Also, we can look at the proportions of the population in each state after one day. The probability vector for the model after 24 hours is:

$$X = \begin{bmatrix} 0.0047 \\ 0.0031 \\ 0.0052 \\ 0.9869 \end{bmatrix}$$

This means that there is a 0.47% chance that the system will be in the Empty state, a 0.31% chance that the system will be in the Tree state, a 0.52% chance that the system will be in the Fire state, and a 98.69% chance that the model will be in the Burnt state. Although this model separates between Empty and Tree states, both of these groups are unaware of the information, so in total, there is a 0.78% chance that the user will not know the information, compared to 30.3% in the modified UID model. Then, the chance of being in the Fire state is similar to the chance of being in the informed state of the UID model, but that model predicts that 47.9% of the

population will be there. Finally, the Burnt state in this model represents a user that stops spreading the information, or becomes disinterested, but there is a 98.69% chance of ending up there, versus the UID model, which predicts that 21.8% of the population will be disinterested after one day. These results are very different from that of the modified UID model. The matrix model includes an absorption state, so once a user reaches it, they cannot leave. However, in the UID model, the reinfection rate allows users that were once disinterested to move back into the informed population, which results in different distributions of users between the two models.

Since the basic UID model does not include a reinfection rate, it may serve as a better comparison for the matrix model. In the matrix model, there is a 98.69% chance of being in the Burnt state after one day, while the basic UID model predicts 79.6% of the population will be in the Disinterested state. These values are still different, but they are much closer to one another than the predictions from the modified UID model. Also, we can compare the values of the Informed class and Fire states between the two models. Both of these groups represent a user who has the information and is actively spreading it to others, so that tells us what percentage of the population is spreading the information at a specific time. This is the most interesting group since we are looking at modeling information spread. In the basic UID model, the peak of the graph of the Informed class happens at about 0.25, meaning that the highest proportion of the population actively spreading the information at one time is 25%. Similarly in the matrix model, the curve of the Fire state peaks at just below 25 users, which would be slightly less than 25% of the total population. These values are very similar given the other differences that there are between the models. Since there are three classes in the UID model, which uses differential equations, and four states in the matrix model, there is always going to be some differences in behavior, but the main groups of interest for modeling information spread are fairly similar. This

means that the basic UID model and the matrix model actually predict similar outcomes for the proportions of the population that spread the information.

Conclusion

The behavior of three different models that predict information spread were compared, both in the long-term and after 24 hours. The basic UID model predicted that in the long-term, the vast majority of users would be in Disinterested and the remaining part of the population would be in the Unaware class, meaning that no one is in the Informed class. In comparison, after 24 hours, the majority of the population is still in the Disinterested class, but the rest of the population is divided between Unaware and Informed. The UID model with the reinfection rate predicted that no one would end up in the Unaware class and the rest of the population would be evenly split between the Informed and Disinterested classes once the equilibrium value is reached. However, after 24 hours, there was more variation between the distributions among classes. Finally, the matrix model predicted that all users would reach the Burnt state in the long-term, because it is an absorption state, but after just 24 hours, almost all users were there already.

Overall, the predictions of the basic UID and the matrix model were the most similar. They weren't exactly the same, but because neither of them allowed users to leave the Disinterested or Burnt states once they got there, their predictions were closer together than those of the modified UID model. Since the modified UID has the reinfection rate, there are always users moving between the Informed and Disinterested classes, so the users are split between the two in the long-term. Neither type of model necessarily gives a better prediction than the other, but the way that each model is designed has led to these differences.

Moving forwards, there is more that can be done to adapt the models to make better comparisons. First, the parameters could be adjusted to be more similar between the modified UID model and the matrix model. The probabilities of moving between states in the matrix model are much higher than the transmission coefficient in the modified UID model, which is likely part of the reason that the differences between the two models are so extreme. While this may make the difference less pronounced, the structures of the models will always lead to some disparities. However, it would be interesting to consider changing the structure of the matrix model to allow movement between states, instead of ending in a single Burnt state. This would make the model more similar to the modified UID model and would make for more clear comparisons.

Citations

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